

pfSense firewall

The file netgate-installer-v1.1.1-RELEASE-amd64.iso.gz was downloaded from the Netgate Store and extracted with the command `gunzip netgate-installer-v1.1.1-RELEASE-amd64.iso.gz`. 60GB was reserved for the disk size, and 2 GB was allocated for the RAM. The Network adapter was set to Bridged: Connected directly to the physical network, and the Replicate the physical network connection state tick-box was checked. The network adapters 2, 3 and 4 were added and set to vmnet2, vmnet3 and vmnet4 respectively. Default settings were accepted during the installation.

Once pfSense was deployed, Option 1 was selected for Assign Interfaces. The interface em0 was designated as WAN, em1 as LAN, em2 as Optional Interface 1 and em3 as Optional Interface 2. Following the interface assignment, option 2 was selected for Set Interface(s) IP address, and the interfaces were allocated IP addresses as per the table below.

Table 1: IP configurations of pfSense firewall interfaces

Interface type	Interface IP address
WAN	DHCP
LAN	10.0.27.1/24
OPT1	10.0.10.1/24
OPT2	10.0.17.1/24

Consequently, the following entries appeared in the pfSense command line interface.

```
pfSense 2.8.1-RELEASE amd64 20251215-1731
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 8d9a60b0f76c553c71fd

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan)    -> em0 -> v4/DHCP4: 192.168.1.128/24
LAN (lan)    -> em1 -> v4: 10.0.27.1/24
OPT1 (opt1)  -> em2 -> v4: 10.0.10.1/24
OPT2 (opt2)  -> em3 -> v4: 10.0.17.1/24
```

Figure 1: Interface configurations in the pfSense console

The pfSense web interface was accessed from the Ubuntu 16.10 web server virtual machine by typing `http://10.0.27.1` in the browser and logging in using the default username `admin` and password `pfsense`.

In the Firewall > Rules > WAN section of the web interface, the subsequent rules were configured or kept as defaults.

Table 2: Firewall rules for the pfSense WAN interface

Protocol	Source	Port	Destination	Port	Gateway
IPv4*	WAN subnets	*	*	*	*
IPv4 TCP	*	*	*	*	*

The Firewall > Rules > LAN section was customised as follows.

Table 3: Firewall rules for the pfSense LAN interface

Protocol	Source	Port	Destination	Port	Gateway
*	*	*	LAN subnets	80	*
IPv4*	LAN subnets	*	*	*	*
IPv6*	LAN subnets	*	*	*	*

In the Firewall > Rules > OPT1 section, the rules listed below were defined.

Table 4: Firewall rules for the pfSense OPT1 interface

Protocol	Source	Port	Destination	Port	Gateway
IPv4*	OPT2 subnets	*	OPT1 subnets	*	*
IPv4*	OPT1 subnets	*	*	*	*

In the Firewall > Rules > OPT2 section, the following rules were applied.

Table 5: Firewall rules for the pfSense OPT2 interface

Protocol	Source	Port	Destination	Port	Gateway
IPv4*	OPT1 subnets	*	OPT2 subnets	*	*
IPv4*	*	*	*	*	*